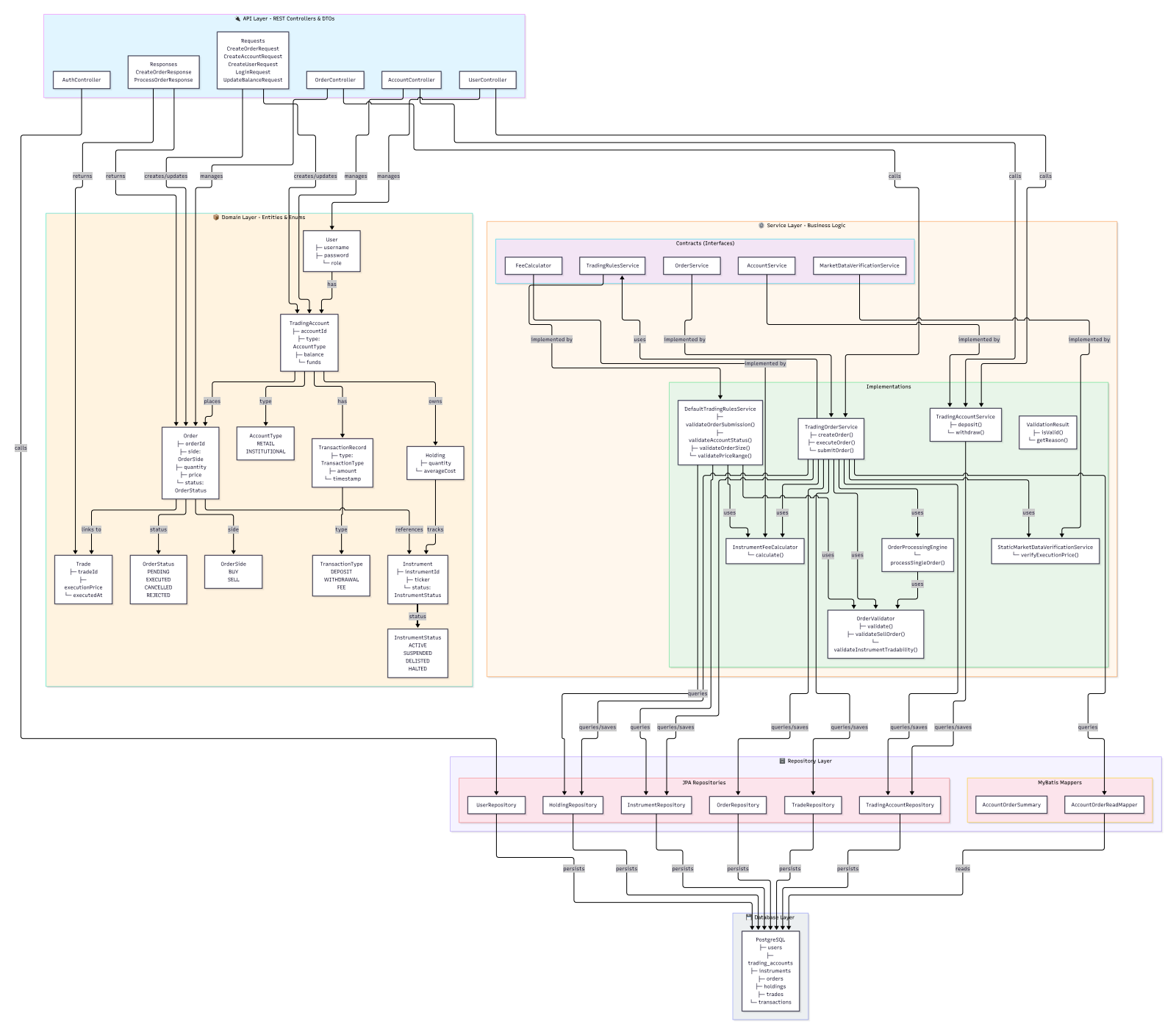

API Layer - REST Controllers & DTOs
AuthController
Responses
CreateOrderResponse
ProcessOrderResponse
Requests
CreateOrderRequest
CreateAccountRequest
CreateUserRequest
LoginRequest
UpdateBalanceRequest
OrderController
AccountController
UserController
returns
returns
creates/updates
manages
creates/updates
manages
manages
calls
calls
calls
calls
Domain Layer - Entities & Enums
User
username
password
role
has
TradingAccount
accountId
type: AccountType
balance
funds
places
type
has
owns
Order
orderId
side: OrderSide
quantity
price
status: OrderStatus
AccountType
RETAIL
INSTITUTIONAL
TransactionRecord
type: TransactionType
amount
timestamp
Holding
quantity
averageCost
calls
links to
status
side
type
references
tracks
Trade
tradeId
executionPrice
executedAt
OrderStatus
PENDING
EXECUTED
CANCELLED
REJECTED
OrderSide
BUY
SELL
TransactionType
DEPOSIT
WITHDRAWAL
FEE
Instrument
instrumentId
ticker
status: InstrumentStatus
status
InstrumentStatus
ACTIVE
SUSPENDED
DELISTED
HALTED
Service Layer - Business Logic
Contracts (Interfaces)
FeeCalculator
TradingRulesService
OrderService
AccountService
MarketDataVerificationService
implemented by
uses
implemented by
implemented by
implemented by
implemented by
Implementations
DefaultTradingRulesService
validateOrderSubmission()
validateAccountStatus()
validateOrderSize()
validatePriceRange()
TradingOrderService
createOrder()
executeOrder()
submitOrder()
TradingAccountService
deposit()
withdraw()
ValidationResult
isValid()
getReason()
uses
uses
uses
uses
InstrumentFeeCalculator
calculate()
uses
uses
OrderProcessingEngine
processSingleOrder()
uses
StaticMarketDataVerificationService
verifyExecutionPrice()
OrderValidator
validate()
validateSellOrder()
validateInstrumentTradability()
queries
queries/saves
queries
queries/saves
queries/saves
queries/saves
queries/saves
queries/saves
queries
Repository Layer
JPA Repositories
UserRepository
HoldingRepository
InstrumentRepository
OrderRepository
TradeRepository
TradingAccountRepository
MyBatis Mappers
AccountOrderSummary
AccountOrderReadMapper
persists
persists
persists
persists
persists
persists
reads
Database Layer
PostgreSQL
users
trading_accounts
instruments
orders
holdings
trades
transactions